SAVITRIBAI PHULE PUNE UNIVERSITY

A MINI PROJECT REPORT ON

“Colorizing Old B&W Images: color old black and white images to colorful images”

Submitted by

CLASS: BE **DIV: A**

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Under the Guidance of

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Academic Year 2025–26(SEM-II)

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CERTIFICATE

This is to certify that the project report entitles

"Colorizing Old B&W Images: color old black and white images to colorful images"

Submitted by

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is a bonafide work carried out by them under the supervision of Prof. Pradnya Kasture and it is submitted towards the partial fulfillment of the requirement of Savitribai Phule Pune University for Final Year.

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Certificate by Guide

This is to certify that Pratik Manjarekar has completed the Mini Project work under my guidance and supervision and that, I have verified the work for its originality in documentation, problem statement, implementation and results presented in the Project. Any reproduction of other necessary work is with the prior permission and has given due ownership and included in the references.

Place : Pune

Date :

Signature of Guide

(Prof. Pradnya Kasture)

Title - Implement Colorizing Old B&W Images: color old black and white images to colorful images

Project title: Colorizing Old B&W Images using CNN

Objective: To build a deep neural network model that can Colorizing Old B&W Images: color old black and white images to colorful images using CNN

Theory - Colorizing black and white images to colorful images involves a complex process that requires expertise and specialized software. However, here are some general steps involved in the process:

Scan the black and white image: The first step is to scan the black and white image and convert it into a digital format.

Preprocess the image: The image needs to be preprocessed to remove any scratches, dust, or other defects. This can be done using image editing software like Photoshop or GIMP.

Convert the image to grayscale: The black and white image needs to be converted to grayscale. This can be done using image editing software or programming languages like Python.

Collect training data: The next step is to collect training data for the colorization model. This can include a dataset of colorful images with their corresponding grayscale versions.

Train the colorization model: A deep learning model can be trained to colorize grayscale images using a dataset of colorful images. This model can be trained using software like TensorFlow, PyTorch, or Keras.

Apply the colorization model to the black and white image: Once the model is trained, it can be applied to the black and white image to generate a colorized version. This can be done using programming languages like Python.

Refine the colorized image: The colorized image may need some manual refinement to ensure that the colors are accurate and the image looks natural. This can be done using image editing software like Photoshop or GIMP.

Save the final image: Once the image has been colorized and refined, it can be saved in a digital format for printing or online use.

Note that the quality of the colorized image will depend on the quality of the original black and white image, the accuracy of the colorization model, and the manual refinement process.

Source Code-

```
import tensorflow as tf
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Conv2D, UpSampling2D, InputLayer,
BatchNormalization
from tensorflow.keras.preprocessing.image import ImageDataGenerator, array_to_img,
img_to_array,
load_img
import numpy as np
# Load the grayscale image
img_gray = load_img('bw_image.jpg', grayscale=True)
# Resize the image to the desired size
img_gray = img_gray.resize((256,256))
# Convert the image to a numpy array
img_gray = img_to_array(img_gray)
# Normalize the image
img_gray = img_gray / 255.0
# Add a new dimension to the array to make it compatible with the input shape of the model
img_gray = np.expand_dims(img_gray, axis=0)
# Load the pre-trained model
model = Sequential()
model.add(InputLayer(input_shape=(None, None, 1)))
model.add(Conv2D(64, (3,3), activation='relu', padding='same', strides=2))
model.add(Conv2D(128, (3,3), activation='relu', padding='same'))
model.add(Conv2D(128, (3,3), activation='relu', padding='same', strides=2))
model.add(Conv2D(256, (3,3), activation='relu', padding='same'))
model.add(Conv2D(256, (3,3), activation='relu', padding='same', strides=2))
model.add(Conv2D(512, (3,3), activation='relu', padding='same'))
model.add(Conv2D(512, (3,3), activation='relu', padding='same'))
model.add(Conv2D(256, (3,3), activation='relu', padding='same'))
model.add(UpSampling2D((2,2)))
model.add(Conv2D(128, (3,3), activation='relu', padding='same'))
model.add(UpSampling2D((2,2)))
model.add(Conv2D(64, (3,3), activation='relu', padding='same'))
```

```
model.add(Conv2D(32, (3,3), activation='relu', padding='same'))
model.add(Conv2D(2, (3, 3), activation='tanh', padding='same'))
model.add(UpSampling2D((2, 2)))
model.compile(optimizer='adam', loss='mse')
# Load the pre-trained weights
model.load_weights('colorization_weights.h5')
# Colorize the grayscale image
img_colorized = model.predict(img_gray)
# Save the colorized image
img_colorized = img_colorized * 128 + 128
img_colorized = np.clip(img_colorized, 0, 255).astype('uint8')
img_colorized = array_to_img(img_colorized[0])
img_colorized.save('colorized_image.jpg')
```

Conclusion- In this way coloring B & W images Implemented.